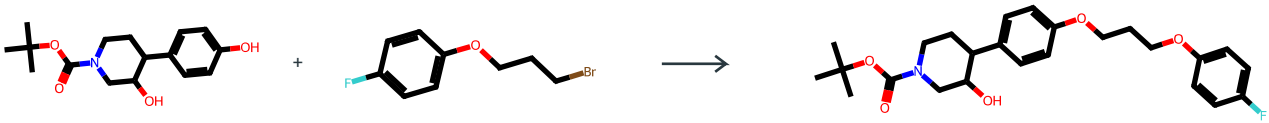
Performed experimental precedent for generated complete reactions

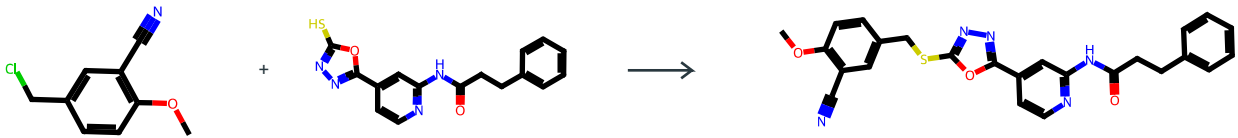
Selected cases; complete net chemistry, not experimentally established electron-arrow sequences.

1. Tier A · US7842688B2:example159c:selective-phenol-alkylation



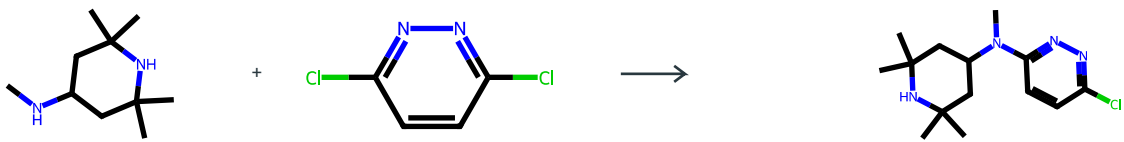
Phenol 0.750 g and bromide 0.758 g, by General Method I: Cs2CO3 in acetonitrile, 80 C for 2 h; aqueous workup and silica chromatography. Slightly yellowish oil, Rf 0.13 (EtOAc/heptane 1:2), Rt 5.12. No isolated mass or percentage yield stated for 159(c).

2. Tier A · US8492378B2:example217:oxadiazole-S-benzylation



Named mercapto-oxadiazolyl pyridyl propionamide and 5-(chloromethyl)-2-methoxybenzonitrile, by Example 1: potassium carbonate in DMF at room temperature for 5 h, workup/chromatography and crystallization. Example 217 reports 39%, colorless crystals, mp 176-177 C, NMR and elemental analysis.

3. Tier A · US8729263B2:preparation1:exocyclic-amine-SNAr



3,6-Dichloropyridazine 4 g/26.8 mmol and N,2,2,6,6-pentamethylpiperidin-4-amine 7.32 g/43.0 mmol in butan-1-ol 67 mL; 120 C for 72 h. Workup, EtOAc trituration and filtration give 4.18 g/14.78 mmol, 55%, off-white solid; MS/NMR.

Source/training overlap unknown; an experimentally documented transformation is not a tested whole route.